MIL-R-93D
SUPPLEMENT 1A
12 June 1970
SUPERSEDING
SUPPLEMENT 1
28 December 1965

MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (ACCURATE),

GENERAL SPECIFICATION FOR

This supplement forms a part of Military Specification MIL-R-93D, 28 December 1965, and is mandatory for use by all Departments and Agencies of the Department of Defense.

DETAIL SPECIFICATIONS

MIL-R-93/3 - Resistors, Fixed, Wire-wound (Accurate), Style RB16.
MIL-R-93/4 - Resistors, Fixed, Wire-wound (Accurate), Style RB17.
MIL-R-93/5 - Resistors, Fixed, Wire-wound (Accurate), Style RB18.
MIL-R-93/6 - Resistors, Fixed, Wire-wound (Accurate), Style RB19.
MIL-R-93/7(EL)- Resistors, Fixed, Wire-wound (Accurate), Style RB43.
MIL-R-93/8(EL)- Resistors, Fixed, Wire-wound (Accurate), Style RB44.
MIL-R-93/9(EL)- Resistors, Fixed, Wire-wound (Accurate), Style RB45.
MIL-R-93/10 - Resistors, Fixed, Wire-wound (Accurate), Style RB52.
MIL-R-93/11 - Resistors, Fixed, Wire-wound (Accurate), Style RB53.
MIL-R-93/12 - Resistors, Fixed, Wire-wound (Accurate), Style RB54.
MIL-R-93/13 - Resistors, Fixed, Wire-wound (Accurate), Style RB55.
MIL-R-93/14 - Resistors, Fixed, Wire-wound (Accurate), Style RB56.
MIL-R-93/15 - Resistors, Fixed, Wire-wound (Accurate), Style RB08.
MIL-R-93/16 - Resistors, Fixed, Wire-wound (Accurate), Style RB57.
MIL-R-93/17 - Resistors, Fixed, Wire-wound (Accurate), Style RB58.
MIL-R-93/18 - Resistors, Fixed, Wire-wound (Accurate), Style RB59.
MIL-R-93/19 - Resistors, Fixed, Wire-wound (Accurate), Style RB70.
MIL-R-93/20 - Resistors, Fixed, Wire-wound (Accurate), Style RB71.
MIL-R-93/27 - Resistors, Fixed, Wire-wound (Accurate), Style RB72.
MIL-R-93/28 - Resistors, Fixed, Wire-wound (Accurate), Style RB73.

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5905-0725)

FSC 5905

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TABLE I. Performance characteristics.

Features	STYLE 1/																						
	RB16				RB17	RB18				RB19				RB52	RB53	RB54	RB55	RB56	RB08	RB57	RB58	RB59	RB70	RB71	RB72	RB73
Resistance tolerance (percent ±) 2/	.05	.1	.5	1	See RB16	.05	.1	.5	1	.05	.1	.5	1	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16
Minimum resistance (ohms)	100	10	.1	.1	See RB16	100	10	1.0	1.0	100	10	5.11	5.11	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16	See RB16
Maximum resistance (.001" dia wire) (megohms)	1.0				2.0	3.01				6.04				1.5	.750	.511	.226	.127	.511	2.0	3.01	5.11	301	.100	3	5
Resistance-temperature characteristic E (ppm/° C (ref 25° C):																										
Under 1 ohm	±90				±90	±90				±90				±90	±90	±90	±90	±90	±90	±90	±90	±90	±90	±90	±90	±90
1 to 10 ohms	±50				±50	±50				±50				±50	±50	±50	±50	±50	±50	±50	±50	±50	±50	±50	±50	±50
10 ohms to max	±20				±20	±20				±20				±20	±20	±20	±20	±20	±20	±20	±20	±20	±20	±20	±20	±20
Power rating (watt) at 125° C	1/3				1/2	3/4				1.0				1/2	1/3	1/4	0.15	1/8	1/4	3/4	1.0	1 1/4	1/4	1/8	1 1/4	1 1/2
Maximum operating voltage (volts)	300				600	600				900				600	300	300	---	---	300	500	900	1,200	150	---	900	1,200
Maximum percent change in resistance 2/																										
Life (full load at 125° C) (1,500 hr for qualification)	±0.5				±0.5	±0.5				±0.5				±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5
Short-time overload	±0.1				±0.1	±0.1				±0.1				±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1
Temperature cycling	±0.2				±0.2	±0.2				±0.2				±0.2	±0.2	±0.2	±0.2	±0.2	±0.2	±0.2	±0.2	±0.2	±0.2	±0.2	±0.2	±0.2	±0.2
Moisture resistance	±0.25				±0.25	±0.25				±0.25				±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±1.0	±1.0	±0.25	±0.25	±0.25
Dielectric withstanding voltage	±0.05				±0.05	±0.05				±0.05				±0.05	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25
Terminal strength (4-1/2 lb pull test) (3-360° rotations)	±0.05				±0.05	±0.05				±0.05				±0.05	±0.05	±0.05	±0.05	±0.05	±0.05	±0.05	±0.05	±0.05	±0.05	±0.05	±0.05	±0.05	±0.05
Salt water immersion cycling	±0.25				±0.25	±0.25				±0.25				±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25
Low temperature operation	±0.25				±0.25	±0.25				±0.25				±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25	±0.25
Low temperature storage	±0.20				±0.20	±0.20				±0.20				±0.20	±0.20	±0.20	±0.20	±0.20	±0.20	±0.20	±0.20	±0.20	±0.20	±0.20	±0.20	±0.20	±0.20
High temperature exposure	±0.50				±0.50	±0.50				±0.50				±0.50	±0.50	±0.50	±0.50	±0.50	±0.50	±0.50	±0.50	±0.50	±0.50	±0.50	±0.50	±0.50	±0.50
Shock	±0.10				±0.10	±0.10				±0.10				±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10
Vibration, high frequency	±0.10				±0.10	±0.10				±0.10				±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10	±0.10
Insulation resistance (megohms):																										
Before moisture resistance	1,000				1,000	1,000				1,000				1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
After moisture resistance	100				100	100				100				100	100	100	100	100	100	100	100	100	100	100	100	100

NOTE: All "C" terminals are solderable in accordance with Method 208 of MIL-STD-202.

1/ See MIL-R-93/7, /8, and /9 for performance characteristics for styles RB43, RB44, and RB45.
2/ Where total resistance change is 1 percent or less, it shall be considered as ± (percent + 0.05 ohm).

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